WxWidgets-VTX

Migration from Wx Version 2.8 to 3.1

WxWidgets Installation

Windows 10

1. Get cygwin64 build tools
    - Browse to: https://www.cygwin.com/
    - Download: setup-x86_64.exe
    - Run setup-x86_64.exe
        - keep default install directories (C:\cygwin64,C:\cygwin-downloads-64)
    - Select Install from Internet
    - Install (at least) the following from Devel tab
        - Basic libraries and build tools
            - binutils,bison,flex,gcc-core,gcc-g++,git,make
        - MinGW Cross-compiler
            - mingw64-x86_64*
    - Close cygwin setup
        - Check add icon to desktop option
2. Open cygwin64 “bash shell” from desktop (double-click on icon on desktop)
3. cd to desired wxWidgets install *parent* directory
    - e.g. $ cd /cygdrive/c

Obtain source code from github

1. Checkout latest wxWidgets from github
    - $ git clone https://github.com/wxWidgets/wxWidgets
2. Set Branch to version 3.1.1
    - $ git checkout tags/v3.1.1 -b v3.1.1
    - note: On 7/6/16 this was the same as the master branch

wxWidgets Build

Windows 10 (64 bit build)

1. Setup build directory
    - In cygwin64 bash shell, cd to wxWidgets install directory

        e.g. $ cd /cygdrive/c/wxWidgets
    - $ mkdir release && cd release
    - set LDFLAGS variable (fixes problem with missing dll at runtime)

        $ export LDFLAGS='-static-libstdc++ -static-libgcc'
2. run configure with the following options (with minGW cross-compiler)
    - ../configure --with-msw --disable-shared --disable-unicode --host=x86_64-w64-mingw32 --build=i686-pc-cygwin --enable-monolithic --with-opengl –disable-precomp-headers
    - note: “--disable-unicode” allows wxString.ToAscii() to return a char* vs. a string buffer which avoids numerable compile errors in current VTX code
    - add -fpermissive to compile flags to remove warnings about const char* to char* casts

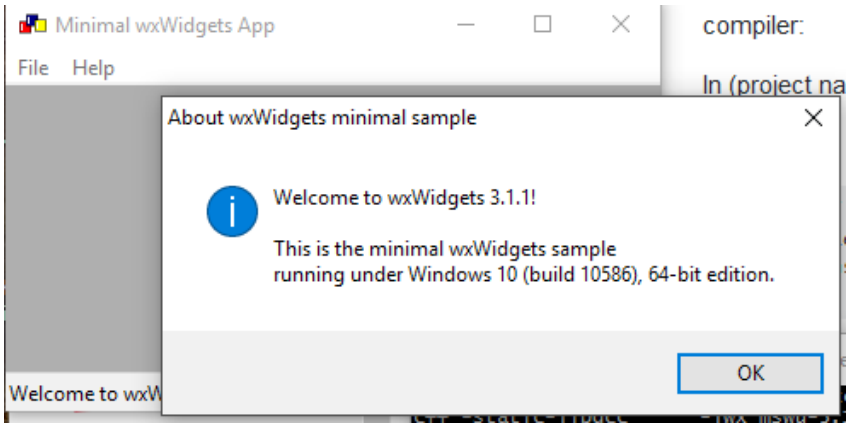
Ubuntu18 (64 bit build)

1. run configure with the following options
    - ../configure --with-gtk --with-opengl --disable-shared --disable-unicode --disable-infobar
        - with –disable-unicode InfoBar.cpp returns an error (breaks build) unless –disable-infobar is also added

VTX Build

3. Build and Install
    - $ make -j8
    - $ make install

4. Test build
    - $ cd samples/minimal
    - $ make
    - $ ./minimal.exe



Ubuntu 14

Setup VTX environment for Eclipse managed build

Define System Environment variables

Windows 10 (Control Panel→System→Advanced->Environment)

1. PLATFORM WIN7
2. GCCBUILD -D_WIN32
3. CYGWIN_HOME C:\cygwin64
4. PATH ${CYGWIN_HOME}/bin;${WXWIN};/usr/local/bin:/usr/bin
5. WXWIN <path-to-wxWidgets-build-directory>
   e.g. C:\wxWidgets\release

Ubuntu (using exports in .bashrc or .profile)

1. PLATFORM LINUX
2. WXVERSION -DWX3

Eclipse Setup

Common

1. While fixing migration problems use a new configuration (e.g WX3)
    - This can be deleted later when all code has been migrated

2. Use `#if wxCHECK_VERSION(3, 0, 0)` statements in code to migrate from 2.8 (default) to 3.1
    - These can be deleted later when all code has been migrated and tested

Project Compile and link flags

Windows 10

1. Toolchain
    - project→properties→settings→C++build→Tool Chain editor
        - Cygwin GCC
2. C++ flags
    - project→properties→settings→C++build→Cygwin C++Compiler
        - Command
            - x86_64-w64-mingw32-g++
        - Command pattern:
            - ${COMMAND} `wx-config --cxxflags` ${FLAGS} ${OUTPUT_FLAG} ${OUTPUT_PREFIX}${OUTPUT} ${INPUTS}
        - preprocessor:
            - WXWIN
_FILE_OFFSET_BITS=64
${PLATFORM}
GLEW
GLEW_STATIC
3. Linker Flags
    - project→properties→settings→C++build→Cygwin C++ Linker
        - Command
            - x86_64-w64-mingw32-g++
        - Command pattern:
            - ${COMMAND} ${FLAGS} ${OUTPUT_FLAG} ${OUTPUT_PREFIX} ${OUTPUT} ${INPUTS} `wx-config --libs --gl_libs`
            - note: `wx-config --libs --gl_libs` must be last thing otherwise get multiple link errors